

ARUN MEENA

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MCA graduate at **NIT Karnataka** with engineering experience at **OLA Electric's Battery Innovation Center**. Skilled in **ML, full-stack development, and data pipeline engineering** with demonstrated impact in cell manufacturing.

EDUCATION

Master of Computer Application

2023 - 2026

National Institute of Technology Karnataka

8.19 CGPA

Bachelor of Computer Application

2020 - 2023

School of Computer Science and IT - DAVV, Indore

7.29 CGPA

INTERNSHIP

OLA Electric – Battery Innovation Center

Jan 2026 - Present

Engineering Trainee

Bangalore, India

- Contributing to **Lithium-ion cell manufacturing and testing research** at OLA Electric's R&D center, supporting cell fabrication and performance validation workflows.
- Upgrading and maintaining **data automation pipelines** that extract raw lab manufacturing data, perform preprocessing, and generate **analysis-ready datasets and auto-plotted visual insights**, while monitoring pipeline stability and implementing new features per evolving research requirements.
- Developing an **automated defect detection system** using **YOLO models** to identify manufacturing defects in cell CT scans, improving inspection accuracy and reducing manual review effort.

PROJECTS

Patent Scope Analyzer– Multi-Agent Patentability Assessment System (In-progress)

- Designing a **multi-agent system** to analyze R&D reports and assess **patentability scope** — identifying novel, already-filed, and improvement-ready patent candidates.
- Scraped and cleaned patent data from **4 major databases** (USPTO, Lens, and others) to build a structured dataset for prior-art comparison.
- Architected a pipeline where a **summarization agent** condenses lengthy R&D reports before downstream patentability analysis.
- Planning **vector embedding**-based similarity search over the patent dataset to retrieve relevant prior art as context for the LLM.
- Designing a final **LLM agent** to synthesize report summary, extracted patentable elements, and similar patents into a patentability readiness score.

Cell Defect Detection - CT Scan Defect Analysis System | Django, React.js, YOLO, Python

- Built an automated defect detection pipeline that processes **CT scan videos** by breaking them into frames and analyzing each using **YOLO models**.
- Trained custom YOLO models on **2,000+** defect images with synthetic data augmentation, achieving detection accuracy of **87–92%** across different defect types.
- Implemented **region segmentation** with intersection-based filtering between defect bounding boxes and inspection zones to eliminate out-of-region false positives.
- Built a **real-time defect streaming** system using **WebSockets** with a **deduplication mechanism** to surface only unique defects per frame on the live stream.
- Developed a **React-based interface** backed by a **Django API** to visualize live defect streams and per-frame detection results.

Cell Test Pipeline – Lab Data Processing & Monitoring System | Django, Celery, Redis, React.js

- Built an **end-to-end automated pipeline** to scan **Google Drive** (mounted via rclone), match directories using **regex**, and extract multi-job testing data from Excel files for **Lithium-ion cell testing teams**.
- Designed **8 job-specific processor functions** that parse raw cell test data, compute key parameters, store calculated metrics in a **database**, and persist plot-ready data as **Parquet files**.
- Implemented a **job status tracking system** with 5 states — Success, In Progress, Failed, Warning (missing file), and Pending Retry (incremental data) — enabling transparent pipeline observability.
- Resolved critical **scalability bottleneck** of **10,000+** jobs by integrating **Celery** with **Redis-backed queues**, enabling parallel worker execution across job types and reducing overall processing time by **more than 75%**.
- Developed a **React dashboard** for the cell testing team to monitor job metrics, statuses, and filter jobs by type, status, and date in real time.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Java, SQL, C

Machine Learning/AI: YOLO, Computer Vision, LLM Agents, Vector Embeddings

Frontend: React.js, Redux, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, Django, Celery, Redis, WebSockets

Databases: PostgreSQL, MySQL, MongoDB

Developer Tools: Git, GitHub, Postman, Linux virtual machine

ACHIEVEMENTS

- Secured **All India Rank (AIR) 150** in **NIMCET 2023**, placing in the **top 0.5%** nationwide and demonstrating proficiency in **Mathematics, Logical Reasoning, Computer Fundamentals, and English**.